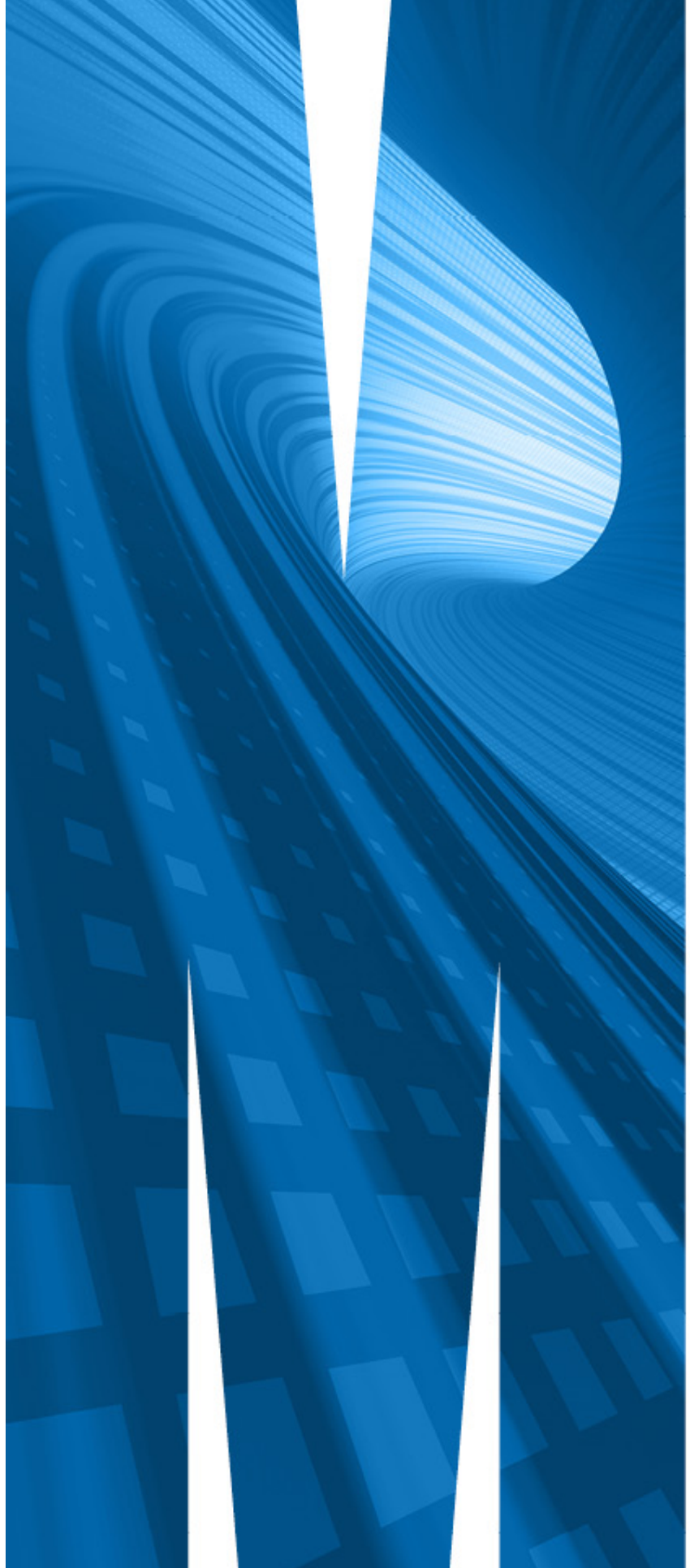
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FIT3179 Data Visualisation

Interactivity



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Ben Shneiderman's mantra:
"*Overview first, zoom and filter, then details-on-demand.*"

Read the short article by Stephen Few, 2006, The Surest Path to Visual Discovery, available on Moodle.

The Surest Path to Visual Discovery
QUESTIONS

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- What are the main purposes of using visual representations?
- What is the greatest benefit of data visualisation?
- What is Ben Shneiderman's mantra, and what does it mean?
- **Overview first**
- **Zoom and filter**
- **Details on demand**
- Is the decision-making process described cyclical or linear?

Overview First

Having an overview is very important. It reduces search, allows the detection of overall patterns, and aids the user in choosing the next move. A good practice in visualisation...is to **start with an overview**.

It displays the closing prices of 1,430 individual stocks across 52 weeks of time.
1) It is clutter. But…
2) this allows you to discern what appears to constitute **normal activity** (closing prices between $7 and $80)
3) as well as exceptions to the norm. **Peak of activity** around week 10.

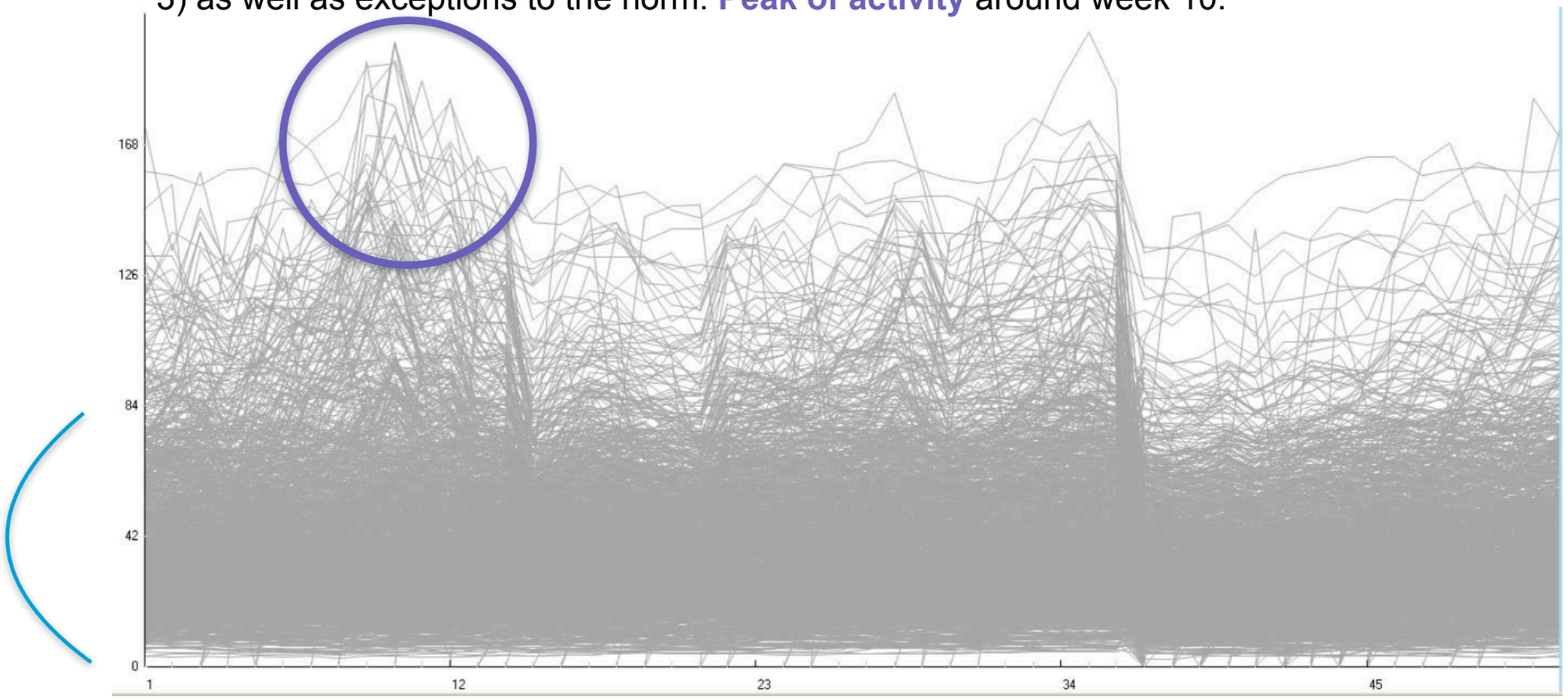


Figure 1: This graph contains a line for each of 1,430 stocks. Each line visually encodes 52 weeks of closing prices.

Source: Stephen Few, 2006, The Surest Path to Visual Discovery.

Zoom…

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Zoom in: … 1) Prices around **week 10** to take a closer look
2) Zoom: narrow the range of time from the full 52 weeks to **weeks 8 through 12**.
3) Benefit: observe the four-week focus and see how it fits into the larger 52-week whole - **focus plus context**.

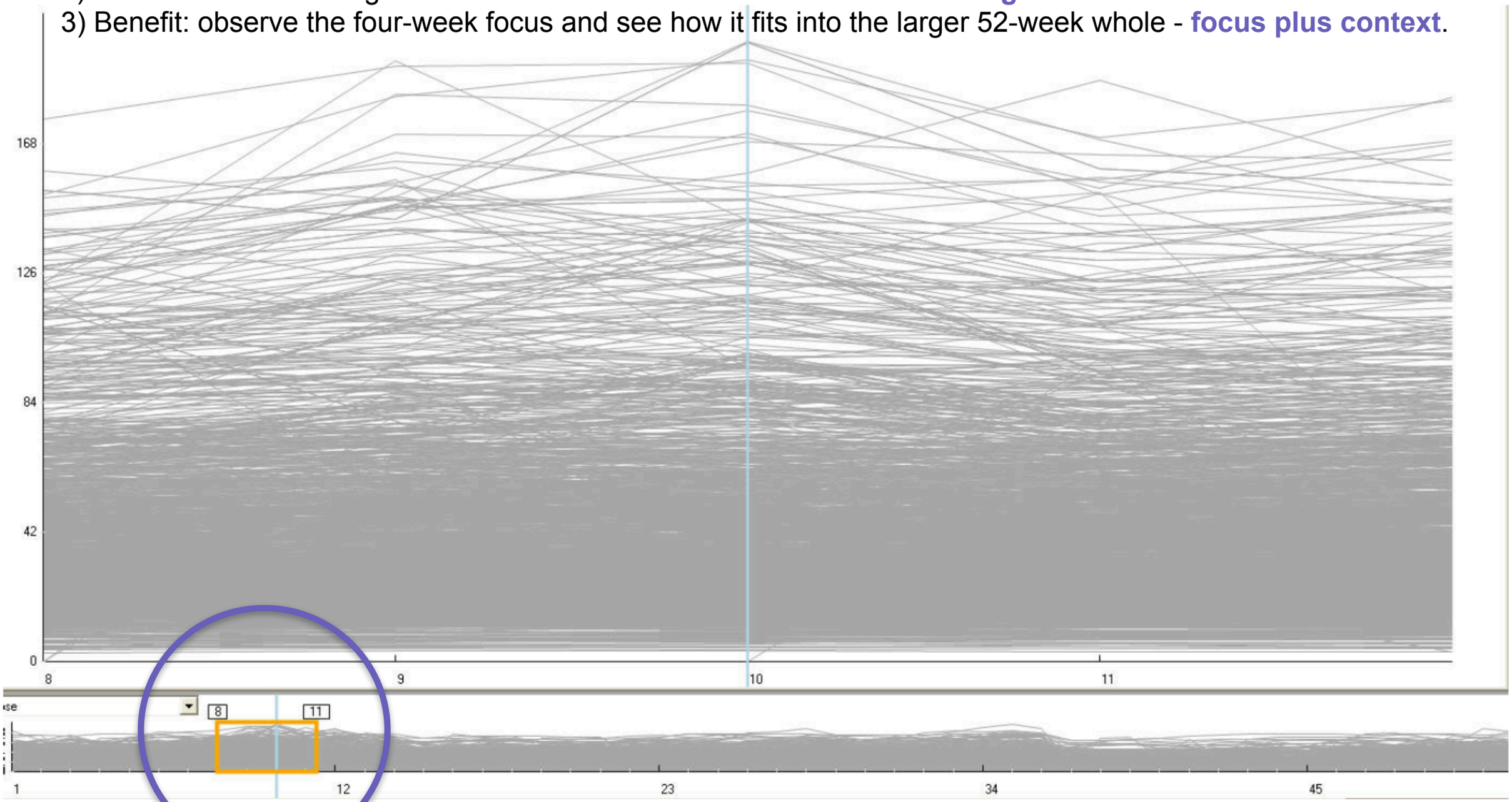


Figure 2: The larger graph above focuses on the four weeks of time that were selected using the gold rectangle in the smaller graph below.

Source: Stephen Few, 2006, The Surest Path to Visual Discovery.

… and filter

1) Explore the **peak of activity around week 10** in greater detail.
2) Filter out distractions (not relevant information)

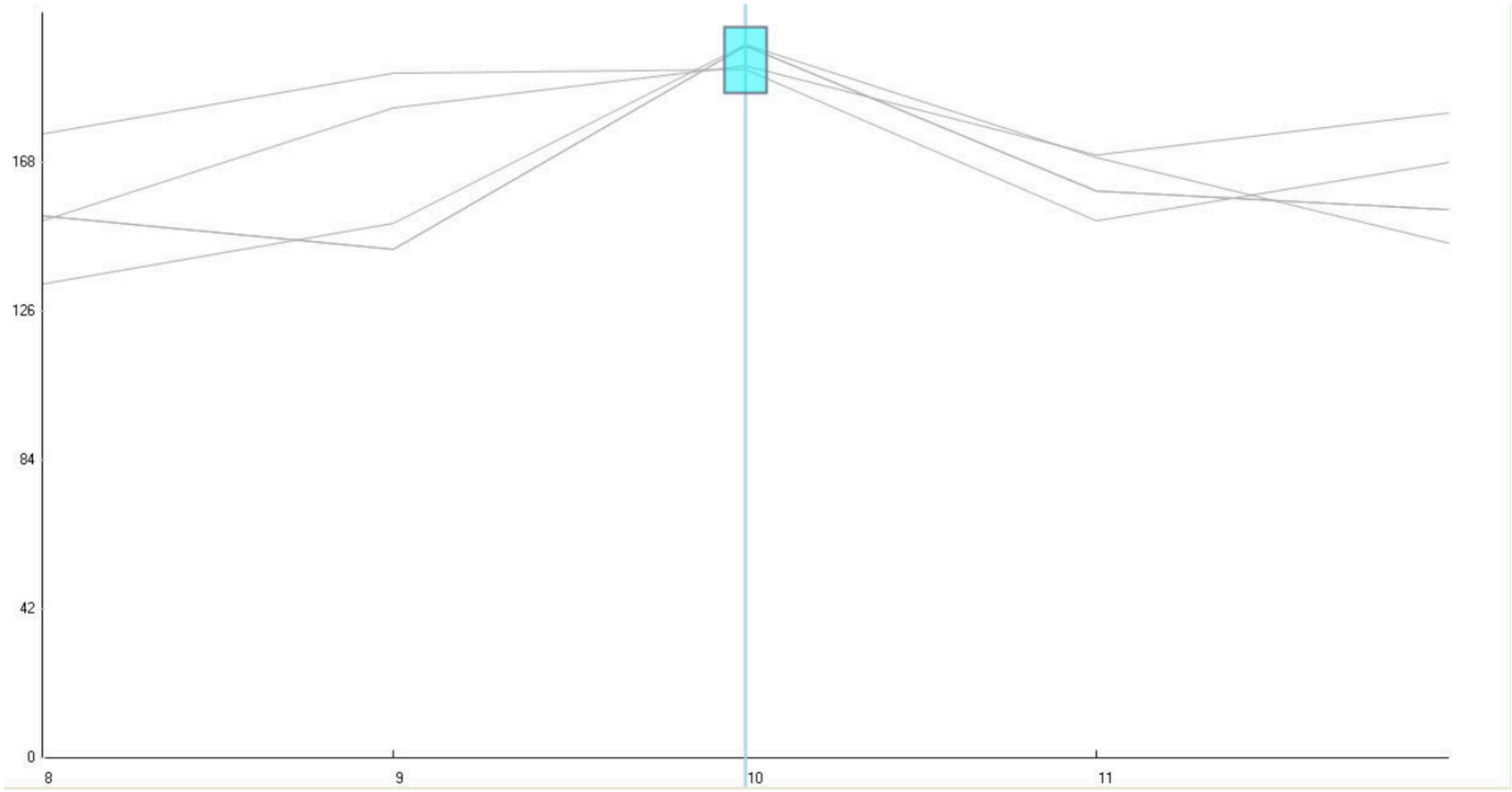


Figure 3: The blue rectangle, a *timebox*, was used to select only those stocks with high closing prices during week 10.

Source: Stephen Few, 2006, The Surest Path to Visual Discovery.

Details-on-demand

1) We may need additional details to understand the pattern, such as **names of these stocks** and their **precise closing prices**.
2) For a particular point in time (**week 11**).
3) Alternatives, hover over particular items will provide additional details.

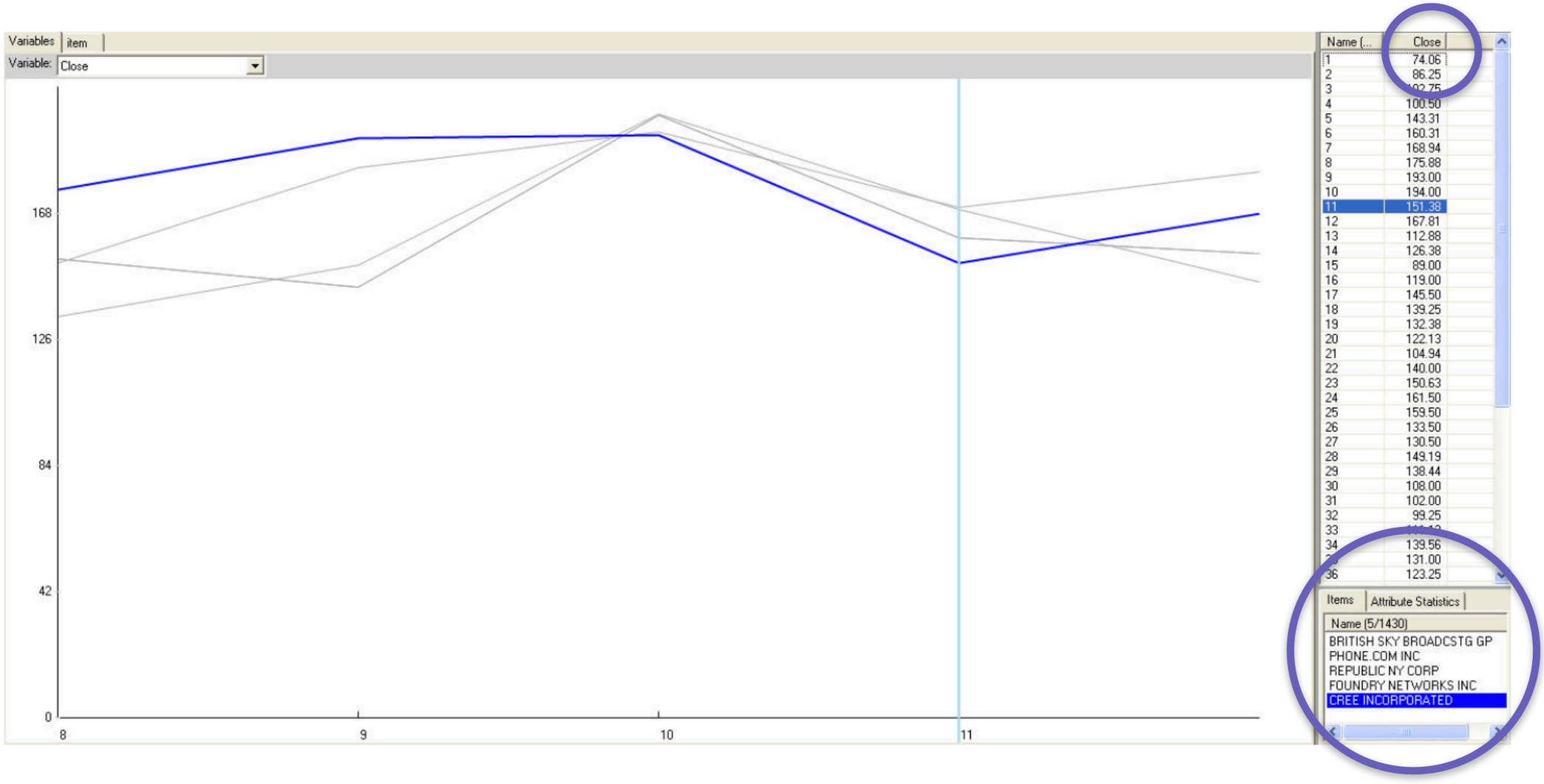


Figure 4: The panels on the right display details as text associated with the line highlighted as a blue line in the graph during week 11.

Source: Stephen Few, 2006, The Surest Path to Visual Discovery.

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Focus+Context

Focus+Context

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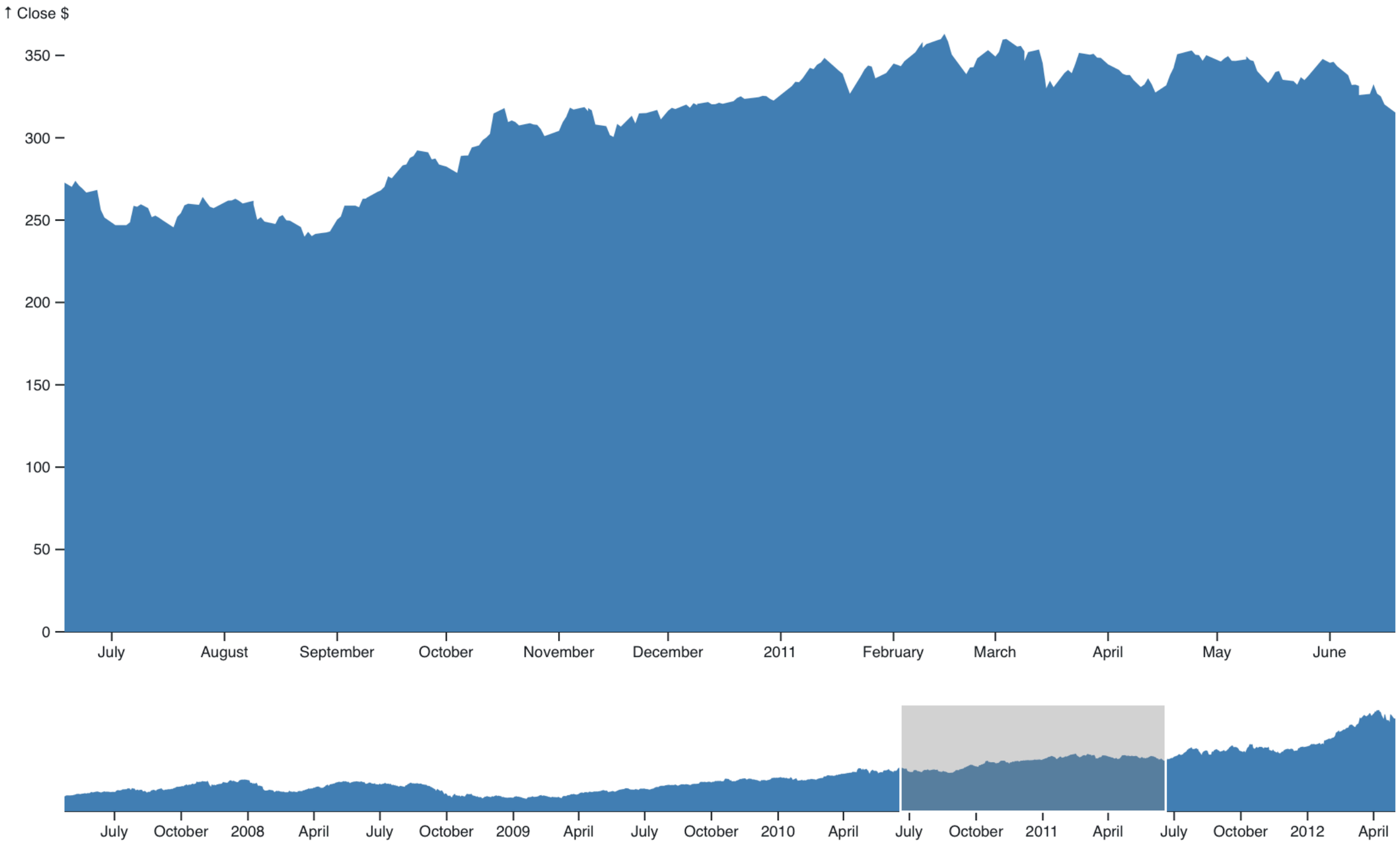
- Show full detail and overview
- Focus = detail view
- Context = surrounding information shown as overview

- Premises
  - User needs both detail (focus) and overview (context) at same time
  - Level of detail is different: focus view with greater detail and context view with reduced detail
  - Combine focus and context in one visualisation

Focus + Context

Not really, because level of detail is almost identical.

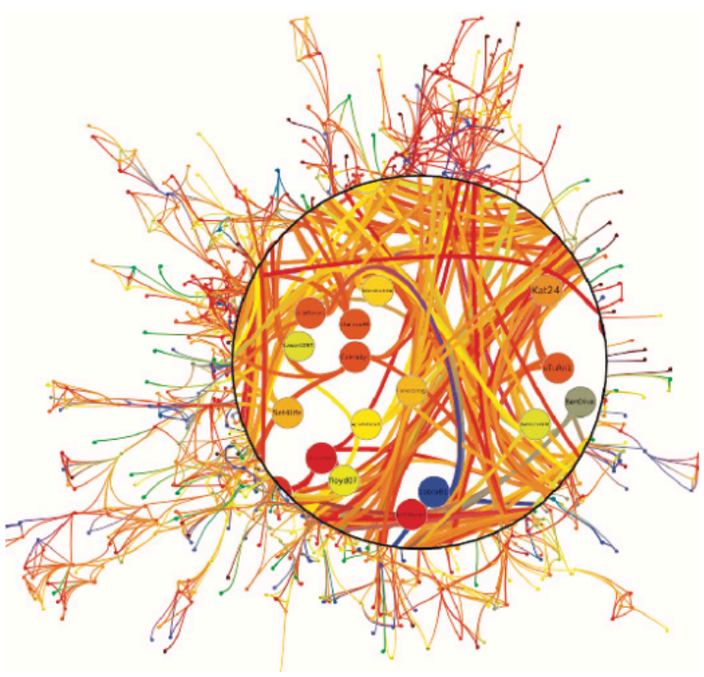
This area chart uses brushing to specify a focused area. Drag the gray region to pan, or brush to zoom. Compare to a zoomable chart. Data: Yahoo Finance



Mike Bostock: https://observablehq.com/@d3/focus-context

Focus+Context





Textbook section 6.5: working memory, cognitive load, animation, change blindness

Activity: Overview first, zoom and filter, then details-on-demand

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Activity 3: Leonardo da Vinci Codex

Which is first? Overview then detail a detailed view then overview

How can you navigate between elements of the data?

...

What data is the overview showing?

...

How can you filter the data shown?

...

How can you zoom in on part of the data?

...

How can you get more details?

...

What aspects rely on short-term memory/long-term memory? How is cognitive load reduced?

...



https://codex-atlanticus.ambrosiana.it/#/Overview

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